Worksheet 2.3 — The Chain Rule

Grade 12 Calculus and Vectors (MCV4U) · Unit 2: Derivative Rules

The Big Ideas

$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$ derivative of the outside (inside untouched) times the derivative of the inside.

- Chain rule: $f'(g(x)) \cdot g'(x)$.
- Power of a function: $\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x)$.
- Don't forget the inner derivative $g'(x)$.

The Big Picture

Differentiating a composition

The **chain rule** handles a function inside a function: $\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$.

Outside times inside

Differentiate the outer function while keeping the inner intact, then multiply by the derivative of the inner.

Layer by layer

For nested functions, apply the chain rule repeatedly, working from the outermost layer inward.

Worked Examples

Example 1 — Power of a function

Differentiate $f(x) = (x^2+1)^3$.

Solution. **Step 1 (identify outer and inner):** outer is $(\cdot)^3$, inner is $g(x) = x^2+1$ (so $g'(x) = 2x$).
Step 2 (differentiate the outer, leaving the inner alone): $3(x^2+1)^2$.
Step 3 (multiply by the inner derivative): $3(x^2+1)^2 \cdot 2x$.
Conclusion: $f'(x) = 6x(x^2+1)^2$.

Example 2 — Radical

Differentiate $f(x) = \sqrt{x^2+1}$.

Solution. **Step 1 (rewrite as a power):** $f(x) = (x^2+1)^{1/2}$, with inner $g(x) = x^2+1$ (so $g'(x) = 2x$).
Step 2 (differentiate the outer power): $\frac{1}{2}(x^2+1)^{-1/2}$.
Step 3 (multiply by the inner derivative): $\frac{1}{2}(x^2+1)^{-1/2} \cdot 2x$.
Step 4 (simplify): the 2's cancel, and $(x^2+1)^{-1/2} = \frac{1}{\sqrt{x^2+1}}$.
Conclusion: $f'(x) = \frac{x}{\sqrt{x^2+1}}$.

Example 3 — Linear inside

Differentiate $f(x) = (3x-2)^5$.

Solution. **Step 1 (identify outer and inner):** outer is $(\cdot)^5$, inner is $g(x) = 3x-2$ (so $g'(x) = 3$).
Step 2 (differentiate the outer): $5(3x-2)^4$.
Step 3 (multiply by the inner derivative): $5(3x-2)^4 \cdot 3$.
Conclusion: $f'(x) = 15(3x-2)^4$.

Example 4 — Cubic inside

Differentiate $f(x) = (x^3+x)^4$.

Solution. **Step 1 (identify outer and inner):** outer is $(\cdot)^4$, inner is $g(x) = x^3+x$ (so $g'(x) = 3x^2+1$).
Step 2 (differentiate the outer): $4(x^3+x)^3$.
Step 3 (multiply by the inner derivative): $4(x^3+x)^3(3x^2+1)$.
Conclusion: $f'(x) = 4(x^3+x)^3(3x^2+1)$.

Example 5 — Chain inside a product

Differentiate $f(x) = x(2x+1)^3$.

Solution. **Step 1 (identify the two factors for the product rule):** $u = x$ (so $u' = 1$) and $v = (2x+1)^3$.
Step 2 (differentiate v with the chain rule): outer $3(\cdot)^2$, inner derivative 2, so $v' = 3(2x+1)^2 \cdot 2 = 6(2x+1)^2$.
Step 3 (apply the product rule $u'v + uv'$): $(1)(2x+1)^3 + x \cdot 6(2x+1)^2$.
Step 4 (factor out the common $(2x+1)^2$): $(2x+1)^2\,[(2x+1)+6x] = (2x+1)^2(8x+1)$.
Conclusion: $f'(x) = (2x+1)^2(8x+1)$.

Example 6 — Power of a function

Differentiate $f(x) = (x^2-4)^3$.

Solution. **Step 1 (identify outer and inner):** outer is $(\cdot)^3$, inner is $g(x) = x^2-4$ (so $g'(x) = 2x$).
Step 2 (differentiate the outer, multiply by the inner derivative): $3(x^2-4)^2 \cdot 2x$.
Conclusion: $f'(x) = 6x(x^2-4)^2$.

Example 7 — Linear inside

Differentiate $f(x) = (5x+1)^4$.

Solution. **Step 1 (identify outer and inner):** outer is $(\cdot)^4$, inner is $g(x) = 5x+1$ (so $g'(x) = 5$).
Step 2 (differentiate the outer, multiply by the inner derivative): $4(5x+1)^3 \cdot 5$.
Conclusion: $f'(x) = 20(5x+1)^3$.

Example 8 — Radical

Differentiate $f(x) = \sqrt{2x+1}$.

Solution. **Step 1 (rewrite as a power):** $f(x) = (2x+1)^{1/2}$, with inner derivative 2.

Step 2 (differentiate the outer, multiply by the inner derivative): $\frac{1}{2}(2x+1)^{-1/2} \cdot 2$.

Step 3 (simplify, the 2's cancel): $(2x+1)^{-1/2}$.

Conclusion: $f'(x) = \dfrac{1}{\sqrt{2x+1}}$.

Example 9 — Reciprocal power

Differentiate $f(x) = \dfrac{1}{(x+1)^2}$.

Solution. **Step 1 (rewrite as a power):** $f(x) = (x+1)^{-2}$, with inner derivative 1.

Step 2 (differentiate the outer, multiply by the inner derivative): $-2(x+1)^{-3} \cdot 1$.

Step 3 (rewrite back as a fraction): $-2(x+1)^{-3} = \dfrac{-2}{(x+1)^3}$.

Conclusion: $f'(x) = \dfrac{-2}{(x+1)^3}$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = (x^2 + 3)^4$.

Question 2

Differentiate $f(x) = (4x - 1)^3$.

Question 3

Differentiate $f(x) = \sqrt{x^2 + 9}$.

Question 4

Differentiate $f(x) = (x^3 - x)^5$.

Question 5

Differentiate $f(x) = \dfrac{1}{(2x + 1)^2}$.

Question 6

Differentiate $f(x) = (1 - x^2)^4$.

Question 7

Differentiate $f(x) = \sqrt{3x - 2}$.

Question 8

Differentiate $f(x) = (x^2 + x + 1)^3$.

Question 9

Differentiate $f(x) = x^2(x + 1)^3$.

Question 10

Differentiate $f(x) = (2x + 5)^6$.

Question 11

Differentiate $f(x) = \dfrac{1}{\sqrt{x + 4}}$.

Question 12

Differentiate $f(x) = (x^2 - 1)^{1/2}$.

Question 13 — Challenge

Differentiate $f(x) = \dfrac{(x + 1)^2}{x}$ (chain + quotient).

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = (x^2 + 3)^4$.
2. Differentiate $f(x) = (4x - 1)^3$.
3. Differentiate $f(x) = \sqrt{x^2 + 9}$.
4. Differentiate $f(x) = (x^3 - x)^5$.
5. Differentiate $f(x) = \dfrac{1}{(2x+1)^2}$.
6. Differentiate $f(x) = (1 - x^2)^4$.
7. Differentiate $f(x) = \sqrt{3x - 2}$.
8. Differentiate $f(x) = (x^2 + x + 1)^3$.
9. Differentiate $f(x) = x^2(x + 1)^3$.
10. Differentiate $f(x) = (2x + 5)^6$.
11. Differentiate $f(x) = \dfrac{1}{\sqrt{x+4}}$.
12. Differentiate $f(x) = (x^2 - 1)^{1/2}$.
13. Differentiate $f(x) = \dfrac{(x+1)^2}{x}$ (chain + quotient).

✓ Answer Key

1. $8x(x^2 + 3)^3$
2. $12(4x - 1)^2$
3. $\dfrac{x}{\sqrt{x^2 + 9}}$
4. $5(x^3 - x)^4(3x^2 - 1)$
5. $\dfrac{-4}{(2x+1)^3}$
6. $-8x(1 - x^2)^3$
7. $\dfrac{3}{2\sqrt{3x - 2}}$
8. $3(x^2 + x + 1)^2(2x + 1)$
9. $2x(x + 1)^3 + 3x^2(x + 1)^2$
10. $12(2x + 5)^5$
11. $-\frac{1}{2}(x + 4)^{-3/2}$
12. $\dfrac{x}{\sqrt{x^2 - 1}}$
13. $\dfrac{2x(x+1)\cdot x - (x+1)^2}{x^2} = \dfrac{(x+1)(x-1)}{x^2}$